

Internal Liquidity Management and Local Credit Provision

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Abstract

Using a unique branch-level dataset of Brazilian banks, this paper studies the patterns of internal liquidity management and how these business practices affect bank lending. Our results suggest first that net due to positions increase during times of financial stress, but this increase is driven by domestically-funded banks, in other words, by banks that are relatively isolated from the stress. Second, headquarter cities of banks tend to have negative due to positions implying that these areas lend money internally to other branches in the banking group. This result is consistent with the headquarter locations raising funds abroad or via wholesale markets and then supplying it to its branches. This negative correlation between the due to position and headquarter locality remains during a period of financial stress and is the same for private and government banks. Third, private banks shift their internal funds during a stress period to richer areas. Lastly, we find that internal liquidity management plays an important role for banks' ability to lend, especially for those exposed to financial stress. Taken together, this paper provides the first branch-level evidence of the way that banks ration liquidity both in normal times and in times of stress, and the impact this has on bank lending.

Keywords: Internal liquidity management, Brazil, bank lending

JEL Classification:

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1. Introduction

The wave of financial globalization that started in the 1980s transformed financial markets and institutions around the world. As a result of this trend of financial integration, global banks increased their footprint within their domestic markets and across both emerging and advanced economies. In this process, banks developed different business models to manage the funds raised from external sources (CGFS, 2010). One of those business models relies intensively on the management of liquidity within the banking organization.

This paper studies the patterns of internal liquidity management for large banks in Brazil and how these business practices affect bank lending to non-related borrowers. In particular, we try to answer two questions: How do banks manage liquidity within their organizations after suffering a liquidity shock? And what is the impact of liquidity management within the banking organization on bank lending and the real economy?

To answer these questions, we use a novel dataset with information on the Brazilian banking sector. The main advantage of these data is that they capture the balance sheets of branches that belong to the same banking organization aggregated by municipality. This information is recorded at a monthly frequency, which helps us investigate the effect of liquidity shocks on the aggregate balance sheet of the banking organization and of its local branches. More important, these data include the net lending of branches to other parts of the organization. This allows us to map, at the micro level, the degree of liquidity management that takes place within the organization as external factors change.

We need a second piece of information to answer our questions. More precisely,

we have to find an external shock that affects Brazilian banks' liquidity conditions, without this shock being correlated with the solvency of those banks or the economic activity of the municipalities in which these banks operate. In our particular sample period, the closest shock with these characteristics is the so called "taper tantrum" (Fischer, 2014). In the spring of 2013, the Chairman of the U.S. Federal Reserve announced that the pace of asset purchases that the central bank was conducting at the time would decelerate in the near future. Financial markets reacted strongly and flows moved quickly out of some emerging markets. Brazilian banks were not immune to this shock and they lost roughly \$20 billion in external funding in two quarters. This shock allows us to identify the reaction of banks within Brazil to the change in liquidity conditions and in particular, their adjustment in net lending within their banking organization as a result of the reduction in external financing.

Figure 1 is a flow chart of how internal liquidity management works mechanically. In Municipality 1, the headquarter location will raise external funds, potentially from foreign sources. It will then lend internally to branches in Municipality 2 and Municipality 3 depending on the liquidity needs of the branches in those locations. It is also possible that Municipality 2 and Municipality 3 lend/borrow from each other. Our data allow us to see the intrabank assets and the intrabank liabilities for each bank in each municipality. We, thus, cannot observe whether Municipality 2 is a net lender to Municipality 3 and a net borrower from Municipality 1. We can observe that this bank in Municipality 2 is a net borrower from the overall banking group (Municipality 1 and Municipality 3 in this example). From this information, we calculate a net due to position for each bank in each location. This position is simply the size of intrabank liabilities net of intrabank assets scaled by total assets in that

location. A positive net due to position, then, implies that the bank operating in a specific location has more intrabank liabilities than assets, which means that it is a net borrower from the banking group. Conversely, a negative net due to position implies that the bank operating in a specific location is a net lender to the banking group.

In our first set of tests, we investigate whether banks follow specific liquidity management patterns across their network of branches. Our results suggest first that net due to positions increase during times of financial stress, but this increase is driven by domestically funded banks, in other words, banks relatively isolated from the stress. Second, our results suggest that headquarter cities of banks tend to have negative due to positions implying that these areas lend money internally to other banks in the banking group. This is consistent with the headquarters location raising money abroad and then supplying it to its branches. Third, that this negative correlation between the due to position and headquarter locality remains during a period of financial stress and is the same for private and government banks. Fourth, that private banks shifted their internal funds during a stress period to areas with higher per capita incomes.

We then test whether this impact on internal liquidity management of Brazilian banks had any effects on their lending to non-related customers. We find that banks with more intrabank liabilities tend to have more lending, consistent with the view that this intrabank funding is actually useful in providing additional credit to the real economy and risk sharing within the banking organization. This effect becomes significantly larger following the start of the taper for those banks particularly exposed to the Fed's taper. In future work we will examine the impact that liquidity

management has on real outcomes including employment and wages.

The study of liquidity management within banking organizations and its impact on the lending activities that these banks conduct has been an active field of research in recent years. Starting with the work of Campello (2002), several papers in the literature have found that risk sharing within banking organizations helps mitigate external shocks, such as changes in monetary policy. This is particularly true for banks that have a large global footprint that allows them to move funds between countries that face different sets of uncorrelated shocks (Cetorelli and Goldberg, 2012).

Another strand of the literature focuses on the real effects of having banking sectors with more geographically diversified banks (Morgan and E., 2004). This literature finds that as bank linkages across regions increase, the fluctuations in the business cycles of those states decrease, but at the same time, the fluctuations of these regions tend to converge.

Similar evidence exists in the economic development literature of risk-sharing across households, where a households' consumption varies less with its own income than with the average income of other households in its village, caste, or ethnic group. Townsend (1994), for example, finds that while income is highly variable across households within Indian villages, consumption is not, with households reallocating income to equalize the marginal utility of consumption. Udry (1994) examines the channels by which such reallocations occur. In his study of households in Northern Nigeria, he finds that households extensively borrow and lend to one another. When a household experiences a negative shock, it will demand payments on loans that it has made while delaying payments on its debts, and thereby smooths income and consumption.

This paper is related to these three strands of the literature, as we examine whether

lending by bank branches within bank networks varies more depending on their own deposits or on the deposit base of their parent banks. We further examine the channels by which any smoothing in lending occurs. Namely, we can observe the interbranch transfers within a bank to determine whether branches are obtaining resources from their branch network or lending resources to their branch network.

2. Brazilian Banking Sector

Brazil’s modern banking history dates back to the early-1800s, with the establishment of foreign banks and domestic banking houses that helped finance the initial debts of the country. During several early banking crises, both national and state governments acted as “insurers against failure” (Musacchio and Lazzarini, 2014) and assumed control of troubled, private banks. The role of Brazil’s government banks expanded during the twentieth century to include the promotion of state-level development projects, the generation of employment, and the distribution of patronage (Ribeiro and Guimaraes, 1967) (Triner, 2000) (Beck and Summerhill, 2005).

Government banks were so politically valuable that by the 1970s, the federal government owned five of them and every state at the time owned at least one of them.¹ State-owned banks were historically mismanaged. Brazil’s monetary authority

¹The five national banks are Banco do Brasil, which was founded in 1808, served as Brazil’s monetary authority until the creation of the Central Bank of Brazil in 1964, and was officially part of the National Treasury until 1987; Caixa Economica Federal, which was established in 1861 as a savings institution; Banco da Amazonia, founded in 1942 to finance rubber cultivation and later re-organized to provide general banking services to the Amazon region; Banco do Nordeste do Brasil, established in 1952 to provide banking services and promote development in the Northeast region; and Banco Nacional de Desenvolvimento Economico e Social (BNDES), a wholesale development bank founded in 1952 to provide long-term financing to infrastructure and strategic sectors. Banks owned by individual state governments were present in all but two of Brazil’s 27 states (including the Federal District), and these two states were formed only more recently.

intervened 71 times in the state-owned banks of 18 states between 1955 and 1996. Given the history of mismanagement, the Brazilian federal government incentivized the recapitalization and privatization of state-owned banks, beginning in 1996, and now only seven states have these institutions.

The location decisions of state and government bank branches also do not appear to react to changes in localities' economic or social characteristics over time. While the initial entry of government bank branches into a locality likely corresponds to the locality's contemporaneous economic and social circumstances, government bank branches almost never exit a locality. This suggests that while a locality's economic and social characteristics evolve, it is not necessarily the case that its bank branch composition evolves with it.

Given their development objectives, a plausible hypothesis is that government banks might lend differently and allocate intrabank funds in a different manner than private-sector banks, especially during times of crisis. With consolidated data for all of a bank's branches in each municipality in Brazil, we are able to examine how intrabank funds flow across different types of localities. Particularly, we can assess whether government banks differentially capture resources from or lend resources to poorer localities or localities with greater external financial dependence, as measured by their employment composition.

3. Data and Empirical Framework

This section discusses the sample selection, the data, and provides summary statistics.

3.1. Sample

For our analysis, we focus on the period between 2011Q1 and 2014Q4 and divide the sample into a pre- and post-“taper” period. Our “taper” variable takes a value of 1 starting in 2013Q2 when the Federal Reserve’s Federal Open Market Committee’s (FOMC) began publicly discussing plans to scale down its quantitative easing program.

Brazil has 5,565 municipalities, which subdivide the states into smaller administrative entities. Because municipalities split and recombine over time, we collapse municipalities into spatially constant units, which we term “localities.” More specifically, we use municipal borders from 1970 and then further combine municipalities that are part of the same urban agglomeration (metropolitan area). Our final sample includes the 2,214 localities that have at least one bank branch, roughly corresponding to individual labor and credit markets.

Currently, approximately one-third of Brazil’s nearly 20,000 bank branches belong to federal government banks, approximately half to private sector banks, and the remainder to state-government banks. Collectively, state and federal government banks account for approximately 45% of total bank assets in Brazil (Barth and Levine, 2013). Our sample of 28 banks consists of government banks and privately-owned domestic and foreign banks. Wanting to exclude some smaller and economically unimportant banks that could drive the results, we first trim the sample to include only those banks that make up the top 99% of assets in the banking sector. Without any reporting errors, we would expect internal borrowing and lending between branches to equal one another when aggregating across all branches for a given bank. We exclude a small number of banks that are believed to be inaccurate reporters when the difference in

these net positions are nontrivial (greater than 1% of consolidated bank assets).

3.2. Data and Summary Statistics

3.2.1. Data

Due to data limitations, previous research has been unable to provide a robust analysis of intrabank funding and how it is used in times of funding stress. For example, the U.S. Summary of Deposits data include information on branch locations and deposits but does not provide broader balance sheet information at the branch or locality level. We overcome this shortcoming in the literature by using a rich database for Brazilian banks, which includes comprehensive financial statements at various levels of aggregation. We utilize both consolidated bank balance sheets and bank balance sheets disaggregated by municipality, which are published by the Central Bank of Brazil at a monthly frequency. For our analysis, we collapse the data to quarterly averages. In the context of internal liquidity management, the granularity of the data allow us observe how different branches within a banking network shift deposits between each other in response to an external funding shock or changes in local economic conditions.

Figures 2 and 3 show the relationship between the average net due to position in a given locality for each bank plotted against log per capita income (figure 2) and log total lending (figure 3). We see for government banks, there does not appear to be a relationship between the income in a given locality and its average net due to position. There is a strong positive relationship, however, between these variables for private banks. This evidence is consistent with bank branches that are located in poorer areas lending money internally to bank branches, within the same banking organization, to bank branches in richer areas. Figure 3 shows the strongly positive

relationship between the net due to position and bank branch lending. This implies that these internal transfers are actually related to the lending that branches do.

After identifying the link between an external funding shock, intrabank lending, and external credit provision, we can examine how these changes in bank operations translate into effects on real outcomes. To measure the local economic impact of these changes in lending, we utilize the Brazilian yearly employment census, *Relacao Anual de Informacoes Sociais* (RAIS). The RAIS identifies all employees on the payroll of formal sector firms as well as the self-employed who pay into the social security system. The data cover approximately 2.5 million establishments and 36 million workers. Finally, we use information on locality-level GDP and control variables including measures of urbanization, education, income, population, and exports, which all come from Brazil’s Institute of Applied Economic Research (IPEA).

3.2.2. Summary Statistics

Figure 4 shows the Brazilian banking sector CDS spread. During this time period, the U.S. Federal Reserve announced a series of unconventional monetary policies. The figure reveals that the stress in the banking sector increased significantly following the announcement of the decision to taper the Federal Reserve’s unconventional monetary policy.

Table 1 provides initial statistics summarizing the sample of banks, the associated net due to positions, and the banking structure. The average locality in our sample has about 2 and a half bank branches, of which, half of them are owned by government banks and half by private banks. Among the private banks, about two-thirds of the branches are owned by domestic banks rather than foreign banks. Banks in Brazil are heavily deposit funded, consisting of over 81 percent of bank branch liabilities, and

the average return on assets of each branch is equal to about 5 percent. The average bank branch has intrabank assets greater than intrabank liabilities that equals about 14 percent of total assets. Lastly, we can note that there is a significant heterogeneity in the level of bank competition across localities.

Tables 2 and Table 3 show how the average net due to position varies by bank ownership and income level, and bank ownership and before/after the taper tantrum, respectively. On average, government bank branches have intrabank assets less than intrabank liabilities of about 10 percent of total bank branch assets, in low income areas, and about 12 percent of total bank branch assets in high income areas. This is in stark contrast to private banks which have significantly negative net due to positions. This is consistent with the development story of government banks that justifies their existence by helping areas obtain credit and grow than would otherwise be unable. We also observe changes in the net due to positions in the pre and post Taper Tantrum period. The average position for government banks doubled from about 7 percent in the pre-Tantrum period to about 15 percent in the post-Tantrum period. Foreign banks additionally saw a substantial change in their net due to positions which went from negative 13 percent to negative 22 percent. Domestic banks declined from negative 34 percent to negative 44 percent.

3.3. Empirical Framework

This paper aims to understand the impact that bank funding stress has on the intrabank market and how this, in turn, impacts local lending and real economic outcomes. To attribute a causal impact, we use the so-called “Taper tantrum” event when the market began to anticipate the Federal Reserve’s shift away from accommodative monetary policies as an exogenous shock to bank funding conditions in

Brazil. In this section we describe our econometric methodology.

We first are interested in understanding the impact of changes in liquidity provision through the interbranch network following the start of the Federal Reserve’s taper. In our analysis, we treat the taper as an exogenous shock to the ability of banks to access funding in international markets, and thus may require banks to rely more heavily on their branch networks. To test this, we run the following specification:

$$y_{ijt} = \alpha + \beta_1 Post_t + \beta_2 PostXForeignFunded_{ijt} + \delta_i + \theta_t + \epsilon_{ijt} \quad (1)$$

where y_{ijt} is the net due to position, calculated as:

$[intrabankliabilities_{ijt} - intrabankassets_{ijt}] / [totalassets_{ijt}]$ for bank i , in locality j , in quarter t . A positive net due to position implies that a bank branch is a net borrower from other bank branches within the banking organization, and a negative position implies that the bank branch is a net lender from other bank branches. Figure 3, for example, shows the distribution of net borrower and net lender localities for the Bank of Brazil. This regression includes bank fixed effects, δ_i , and we additionally include time fixed effects, θ_t , in alternative specifications. Note that the dummy variable for being a government bank is omitted as it is collinear with the bank fixed effect. In all of our estimations in this and the following sections, we cluster at the bank level.

Because the headquarter office of a bank is likely to have greater access to external funding, we can differentiate bank branches as being either in the headquarter city or not. Figure 4 shows the geographic distribution of the headquarters of banks in Brazil. In particular, our sample’s headquarter locations are largely concentrated in Sao Paulo and Brasilia.

To test how banks changed their liquidity management with respect to its head-

quarter location during the Federal Reserve’s taper, we run the following specification:

$$y_{ijt} = \alpha + \beta_1 Post_t + \beta_2 headquarters_{ij} + \beta_3 PostXheadquarters_{ij} + \delta_i + \sigma_j + \theta_t + \epsilon_{ijt} \quad (2)$$

where again y_{ijt} is the net due to position as calculated above. In these regressions we include either bank and locality fixed effects or bankXlocality fixed effects in alternative specifications. We additionally run regressions where instead of including a dummy for the headquarter locality of a bank, we include an indicator of economic development. In these regressions, we are interested in understanding whether intrabank transfers flow from rich to poor localities or vice-versa.

We ultimately aim to test what impact this shock to bank funding has on lending at the locality level. To test this, we run the following specification:

$$y_{ijt} = \alpha + \dots + \beta_1 PostXNetDueToXForeignFunded_{ijt} + \delta_i + \sigma_j + \theta_t + \epsilon_{ijt} \quad (3)$$

where y_{ijt} is the natural logarithm of total credit operations for bank i in locality j in time t. We include controls at the bank-by-locality level, at the banking group level, and at the locality level. We additionally include the size of the intrabank positions for each branch over time. Our hypothesis is that bank branches will lend more if their intrabank liabilities are higher ($\beta > 0$) because it is precisely these liabilities that will allow them to continue their credit expansion if they run out of deposits to lend. The last test that we run expands equation (3) to include the “Post” dummy to see if these relationships held in the environment with higher bank funding stress. We will be expanding this analysis to see if the differential changes in lending caused by the taper had any impact on local employment, wages, and firm growth.

4. Results

We first present results that show how the net due to positions of bank branches vary following the “Taper tantrum.” We allow for a differential impact on foreign funded and domestically funded banks as foreign funded banks are more exposed to the United States’ financial sector. Instead of simply using a dummy for the “tantrum,” we additionally use the Brazilian bank CDS spread index which will capture the stress of the Brazilian banking sector at any given moment. We then test whether there are differences between branches located in the headquarters city or in wealthy cities to understand whether it is money being transferred from the country side to cities or vice versa. Our last set of results looks at the impact of the net due to position on lending by bank branches in Brazil.

4.1. *Net Due To Position*

Table 4 shows the results from estimating equation (1) for quarterly net due to positions at the bank-by-locality level on a *Post* period dummy for the “Taper tantrum,” and the interaction between *Post* and a dummy variable for whether the bank is foreign funded. The even number columns alternatively report results using a dummy for a bank being privately owned (instead of government owned) which in subsequent tables we use as our proxy for foreign exposure.

Column (1) includes bank and quarter fixed effects and shows that net due to positions increase during the taper tantrum but this effect is driven by domestically funded banks. Column (2) shows the same results but uses private banks as an indicator for foreign exposure as opposed to foreign funding, and we find the same to be true. These results are consistent with banks that are internationally exposed being

less able to fund themselves and thus needing to reduce their intrabank exposures. Columns (3) and (4) show the same results including bank, city, and time fixed effects and we find the same results. Lastly, to control for local demand conditions, we include bank and cityXtime fixed effects in columns (5) and (6) and find similar results.

Instead of using a dummy variable for the “Taper tantrum” to proxy for stress in the Brazilian banking sector as in Table 4, one could instead use a bank-weighted credit default swap spread to provide a more contemporaneous measure of stress. The aggregate CDS index is shown in Figure 4 with a vertical line indicating the beginning of the “Taper tantrum.” We see that following the start of the “tantrum” bank CDS spreads increase significantly, suggesting an increase in banking system stress in Brazil.

Table 5 presents the results of Table 1 but using the Brazilian banking sector CDS spread instead of the “*Post*” dummy. Again, we find positive, and marginally statistically significant coefficients on the CDS spread implying that as banking system stress increases, banks fund themselves through internal liquidity management more prominently. This effect, though, is again driven by the domestically funded banks (in columns (1), (3), and (5)) and through private banks (in columns (2), (4), and (6)).

One innovation of this paper is that we are able to see the net due to position of banks at the locality level. We are thus able to see from where banks are moving funds in times of stress. Are banks moving internal funds from the headquarter location, where they may have been able to obtain funds through capital markets? Are banks moving internal funds from poorer areas with few viable investment opportunities to

richer areas with more plentiful investment uses?

Table 6 reports results of the net due to positions of banks and how they vary by time and international exposure. For the remainder of the analysis we compare private banks with domestic banks as they provide a cleaner proxy for foreign exposure than our constructed measure of foreign funding; the two measures yield the same qualitative results.

We find consistent evidence that there is a negative relationship between being a headquarters of a bank and net due to positions in Brazil. This means that the headquarter city has higher intrabank assets than it has intrabank liabilities implying that headquarters locations manage liquidity through lending money to branches not located in the headquarter city. We do not, however, find that this negative relationship changed during the period of bank stress. This result is consistent whether we include bank, city, and quarter fixed effects, as we do in columns (1)-(3) or bank and locality-by-time fixed effects as we do in columns (4)-(6). Even though we added some additional controls and interactions, the negative effect on private banks during the “Taper tantrum” persists.

Table 7 shows results consistent with Table 6, but using the Brazilian banking sector CDS spreads as the measure of bank stress instead of the dummy for the “Taper tantrum” period. The results overall are similar, namely that headquarter locations are net lenders of internal liquidity, but this did not change markedly during the period of greater bank stress. Private banks decreased their internal liquidity management during this period, but there was no difference in the use of headquarter locations between private and government banks during the period of bank stress.

Our rich dataset also allows us to see whether banks manage internal liquidity

by shifting money from rich to poor areas or vice-versa. Tables 8 and 9 explore this issue. In Table 8, we find that private banks in high income areas, defined as above the median per capita gross municipal product, have higher net due to positions meaning that private banks shift their internal liquidity to richer areas, presumably because investment prospects are higher in these areas. In the regressions with locality-by-time and bank fixed effects, we find that this effect increased, relative to poorer areas, during the stress period. Private banks overall reduced the extent to which funds were re-allocated across branches within their bank network during the stress period, however.

We present the same qualitative picture in Table 9 where we replace the “Post” dummy with the Brazilian bank CDS spreads. The coefficients in this table have the same sign as in Table 8 but with less statistical significance. Tables 10 and 11 report similar results to those in Tables 8 and 9 using a continuous measure of income instead of a dummy for being above or below the median per capita income level, and we find the results still hold.

As a robustness check, we show an alternative to our baseline results reported in Tables 4-11 where we cluster at the bank level. Because the sample size of banks in Brazil is limited, a concern is small sample clustering. In Table 12 columns (1) and (2), we provide results of the net due to positions of banks regressed on “Post,” “PostXForeign Funded,” and “PostXPrivate Bank” and clustered at the bank level which are the same regressions as in Table 4. Columns (3) and (4) replace “Post” with the CDS spreads as in Table 5. Columns (5)-(8) provide bootstrapped results. As expected, the coefficient estimates are identical as we are only adjusting the standard errors, but we additionally find that the significance of our results remains even after

the bootstrapping procedure.

Our internal liquidity management results suggest the following: First, net due to positions increase during times of financial stress, but this increase is driven by domestically funded banks, in other words, by banks that are relatively isolated from the stress. Second, headquarter cities of banks tend to have negative due to positions implying that these areas lend money internally to other banks in the banking group. This is consistent with the headquarter locations raising money abroad or via wholesale markets and then supplying it to its branches. Third, this negative correlation between the due to position and headquarter locality remains during a period of financial stress and is the same for private and government banks. Fourth, private banks shift their internal funds during a stress period to richer areas. These results hold whether we used a dummy for the “Taper tantrum” or the CDS spreads, control for various specifications of fixed effects, or bootstrap standard errors.

4.2. Lending

The previous section showed how the net due to positions of bank branches changed when bank funding became stressed. This section presents results on how lending is related to the net due to positions of banks and then how it was impacted by the taper.

Table 13 presents our first results on how lending is related to the net due to positions of bank branches. In columns (1) and (2) we see that the net due to position is positively correlated with locality level lending which implies that bank branches that are net borrowers are using that money to increase lending beyond what would be possible using only local deposits. We do not find for the overall sample that this dependence on internal liquidity becomes more important during the stress period.

We do, however, find that banks that are more exposed to the international financial system rely more heavily on internal liquidity management for their lending and that this dependence became stronger during the “Taper tantrum.”

5. Discussion and Conclusion

Using a unique dataset that allows us to see the interbranch operations of bank networks, we analyze how banks utilize their intrabank market to raise funding, e.g., take deposits from certain locations and transfer them to other branches within their banking group. As far as we know, this is the first analysis that has been able to use bank balance sheets at the city level which allows us to clearly identify the dynamics of internal bank liquidity provision.

Our internal liquidity management results suggest first that net due to positions increase during times of financial stress, but this increase is driven by domestically funded banks, in other words, banks relatively isolated from the stress. Second, that headquarter cities of banks tend to have negative due to positions implying that these areas lend money internally to other banks in the banking group. This is consistent with the headquarters location raising money abroad and then supplying it its branches. Third, that this negative correlation between the due to position and headquarter locality remains during a period of financial stress and is the same for private and government banks. Fourth, that private banks shifted their internal funds during a stress period to richer areas. These results hold whether we used a dummy for the “Taper tantrum” or the CDS spreads, control for various specifications of fixed effects, or bootstrap standard errors. We additionally find that internal liquidity management plays an important role for banks’ ability to lend, especially for those

exposed to financial stress.

Taken together, this paper provides the first branch-level evidence of the way that banks ration liquidity both in normal and in stressful times and the importance of this for banks to continue lending.

Citations: (Barth and Levine, 2013) (Beck and Summerhill, 2005) (Bernanke and Blinder, 1988) (Cetorelli and Goldberg, 2012) (Coleman and Feler, 2015) (CGFS, 2010) (I.M.F., 2013) (Jimenez and Saurina, 2014) (Kashyap and Stein, 2000) (Khwaja and Mian, 2008) (Morgan and E., 2004) (Musacchio and Lazzarini, 2014) (Ribeiro and Guimaraes, 1967) (Townsend, 1994) (Triner, 2000) (Udry, 1994)

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6. Tables

Table 1: Summary Statistics

	Mean	Median	Stand Dev.	Min	Max	Localities
<u>By Locality:</u>						
<u>Bank Composition:</u>						
Number of Banks	2.44	2	1.92	1	15	2,373
Government Banks	1.32	1	.90	0	7	2,373
Domestic Banks	.79	1	.90	0	6	2,373
Foreign Banks	.34	0	.63	0	2	2,373
Total Branches	8.11	2	76.25	1	3103	2,373
Government Branches	3.50	1	22.70	0	805	2,373
Domestic Branches	3.23	1	38.19	0	1591	2,373
Foreign Branches	1.38	0	16.69	0	707	2,373
<u>Bank Characteristics:</u>						
Net Due To/Total Assets	-.139	-.13	.38	-.90	.89	2,373
Total Credit/Total Assets	.93	.96	.09	.27	.99	2,373
Net Income/Total Assets	.05	.04	.04	-.15	.28	2,373
Deposits/Total Liabilities	.81	.89	.20	.08	.99	2,373
<u>Banking Structure:</u>						
Herfindahl Index: Deposits	.63	.5	.33	.14	1	2,373
Herfindahl Index: Total Assets	.73	.83	.29	.13	1	2,373
<i>Source:</i> Central Bank of Brazil						

Table 2: Summary Statistics: Net Due To/Total Assets

Net Due To/Total Assets	Mean	Median	Stand Dev.	Bank Locations
<u>Below Median Per Capita Income:</u>				
All Banks	-.130	-.056	.445	2,798
Government Banks	.096	-.000	.343	1,677
Domestic Banks	-.517	-.585	.319	994
Foreign Banks	-.111	-.155	.409	127
<u>Above Median Per Capita Income:</u>				
All Banks	-.084	-.018	.454	4,019
Government Banks	.123	-.000	.335	1,853
Domestic Banks	-.300	-.463	.503	1,465
Foreign Banks	-.179	-.231	.366	701

Source: Central Bank of Brazil

Table 3: Summary Statistics: Net Due To/Total Assets

Net Due To/Total Assets	Mean	Median	Stand Dev.	Bank Locations
<u>Pre-period:</u>				
All Banks	-.103	-.043	.440	6,656
Government Banks	.073	-.000	.340	3,379
Domestic Banks	-.339	-.470	.467	2,450
Foreign Banks	-.125	-.179	.384	827
<u>Post-period:</u>				
All Banks	-.110	-.019	.478	6,809
Government Banks	.147	-.000	.344	3,529
Domestic Banks	-.441	-.594	.452	2,453
Foreign Banks	-.224	-.310	.387	827

Source: Central Bank of Brazil

Table 4: Total Intrabank Activity

	(1) Total Activity	(2) Total Activity	(3) Total Activity	(4) Total Activity	(5) Total Activity	(6) Total Activity
Post	19.285*** (4.286)	-4.677 (6.586)	2.505 (6.093)			
Foreign FundedXPost		31.043** (7.960)				
Private BankXPost			22.911*** (7.305)			
CDS Spread				0.171*** (0.053)	-0.051 (0.076)	0.011 (0.069)
Foreign FundedXCDS					0.318*** (0.087)	
Private BankXCDS						0.242*** (0.082)
<i>N</i>	103264	103264	103264	103264	103264	103264
<u>Fixed Effects:</u>						
Bank	✓	✓	✓	✓	✓	✓
City	✓	✓	✓	✓	✓	✓
Quarter	✓	✓	✓	✓	✓	✓

Table 5: Total Intrabank Activity

	(1) Total Activity	(2) Total Activity	(3) Total Activity	(4) Total Activity	(5) Total Activity	(6) Total Activity
Sov. CDS	0.146** (0.062)	-0.017 (0.070)	-0.017 (0.081)			
Foreign FundedXSov. CDS		0.265*** (0.100)				
Private BankXSov. CDS			0.265*** (0.100)			
Ex. Rate				40.143*** (11.597)	-19.495 (20.364)	-19.495 (20.364)
Foreign FundedXEx. Rate					63.433*** (17.692)	
Private BankXEx. Rate						63.433*** (17.692)
<i>N</i>	103264	103264	103264	103264	103264	103264
<u>Fixed Effects:</u>						
Bank	✓	✓	✓	✓	✓	✓
City	✓	✓	✓	✓	✓	✓
Quarter	✓	✓	✓	✓	✓	✓

Table 6: Net Due to Position

	(1)	(2)	(3)	(4)	(5)	(6)
	Net Due To	Net Due To	Net Due To	Net Due To	Net Due To	Net Due To
Post	-0.015*	0.034**	0.037***			
	(0.009)	(0.013)	(0.012)			
Foreign FundedXPost		-0.063***				
		(0.012)				
Private BankXPost			-0.071***			
			(0.012)			
CDS Spread				-0.000	0.000	0.000
				(0.000)	(0.000)	(0.000)
Foreign FundedXCDS					-0.001***	
					(0.000)	
Private BankXCDS						-0.001***
						(0.000)
<i>N</i>	103264	103264	103264	103264	103264	103264
<u>Fixed Effects:</u>						
Bank	✓	✓	✓	✓	✓	✓
City	✓	✓	✓	✓	✓	✓
Quarter	✓	✓	✓	✓	✓	✓

Table 7: Net Due to Position

	(1)	(2)	(3)	(4)	(5)	(6)
	Net Due To	Net Due To	Net Due To	Net Due To	Net Due To	Net Due To
Sov. CDS	-0.000	0.000	0.000			
	(0.000)	(0.000)	(0.000)			
Foreign FundedXSov. CDS		-0.000				
		(0.000)				
Private BankXSov. CDS			-0.000			
			(0.000)			
Ex. Rate				-0.008	0.172***	0.172***
				(0.049)	(0.042)	(0.042)
Foreign FundedXEx. Rate					-0.191***	
					(0.057)	
Private BankXEx. Rate						-0.191***
						(0.057)
<i>R</i> ²	0.89	0.89	0.89	0.89	0.89	0.89
<i>N</i>	103264	103264	103264	103264	103264	103264
<u>Fixed Effects:</u>						
Bank	✓	✓	✓	✓	✓	✓
City	✓	✓	✓	✓	✓	✓
Quarter	✓	✓	✓	✓	✓	✓

Table 8: Total Intrabank Activity

	(1)	(2)	(3)	(4)	(5)	(6)
	Total Activity	Total Activity	Total Activity	Total Activity	Total Activity	Total Activity
Interaction	Headquarters	Income	Urbanization	Industrialization	Concentration	Earnings
Total Activity (lagged)	866.743*** (19.387)	864.982*** (20.928)	866.133*** (19.825)	867.263*** (19.213)	866.791*** (19.384)	869.516*** (18.694)
Post	-2.533 (4.027)	-1.761 (4.137)	11.183*** (3.398)	0.458 (3.914)	-6.621* (3.420)	-2.978 (4.006)
High For. FundedXPost	2.289 (4.766)	4.807 (3.400)	-7.761 (5.818)	-0.121 (3.805)	3.143 (4.899)	4.767 (5.238)
High For. FundedXInteraction	-65.845* (33.542)	-10.290* (5.809)	-31.536* (17.522)	-5.479 (15.735)	14.397 (11.603)	73.710 (62.419)
PostXInteraction	-7.995** (3.558)	-1.672** (0.712)	-19.737*** (3.085)	-16.355** (6.308)	8.189* (3.897)	14.802 (11.450)
High For. FundedXPostX Interaction	4.711 (6.939)	-3.528 (4.325)	14.831 (9.950)	13.390 (9.758)	-11.544 (8.788)	-75.217 (51.983)
R^2	0.95	0.95	0.95	0.95	0.95	0.95
N	96447	96447	96447	96447	96447	96447

Notes:

Table 9: Total Intrabank Activity

	(1)	(2)	(3)	(4)	(5)	(6)
	Total Activity	Total Activity	Total Activity	Total Activity	Total Activity	Total Activity
Post	-4.527 (7.009)	0.555 (6.876)	39.739*** (6.001)	9.526 (7.410)	-6.319 (13.949)	0.419 (9.261)
High For. FundedXPost	30.823** (10.780)	32.414* (17.375)	-11.613 (40.890)	17.169 (13.925)	31.499** (11.954)	7.975 (19.076)
High For. FundedXInteraction	-424.861** (172.858)	-94.973*** (28.611)	-194.289* (102.321)	-63.771 (108.534)	70.903 (76.500)	-732.869* (370.032)
PostXInteraction	-20.196 (15.226)	-9.048*** (2.456)	-63.167*** (6.831)	-75.895*** (16.994)	14.152 (18.800)	-5.911 (9.523)
High For. FundedXPostX Interaction	-29.809 (30.581)	-2.303 (22.260)	59.782 (50.573)	73.530* (41.367)	-67.696** (25.699)	449.994** (173.827)
R^2	0.74	0.74	0.74	0.73	0.74	0.75
N	103264	103264	103264	103264	103264	96447

Notes:

Table 10: Net Due To

	(1)	(2)	(3)	(4)	(5)	(6)
	Net Due To	Net Due To	Net Due To	Net Due To	Net Due To	Net Due To
Interaction	Headquarters	Income	Urbanization	Industrialization	Concentration	Earnings
Net Due To (lagged)	0.778*** (0.046)	0.774*** (0.047)	0.777*** (0.046)	0.778*** (0.046)	0.777*** (0.046)	0.777*** (0.046)
Post	0.013*** (0.004)	0.014*** (0.005)	0.015* (0.007)	0.013** (0.005)	-0.004 (0.005)	0.011** (0.004)
High For. FundedXPost	-0.004 (0.006)	0.004 (0.006)	0.035*** (0.007)	0.001 (0.006)	-0.001 (0.005)	0.009 (0.009)
High For. FundedXInteraction	0.015 (0.024)	0.042*** (0.012)	0.096*** (0.025)	0.040 (0.027)	-0.045* (0.022)	0.409* (0.202)
PostXInteraction	-0.007 (0.007)	-0.004 (0.003)	-0.004 (0.009)	0.000 (0.015)	0.026** (0.011)	0.023** (0.010)
High For. FundedXPostX Interaction	-0.015 (0.009)	-0.012*** (0.003)	-0.049*** (0.010)	-0.022 (0.017)	0.001 (0.017)	-0.333** (0.131)
R^2	0.97	0.97	0.97	0.97	0.97	0.97
N	96447	96447	96447	96447	96447	96447

Notes:

Table 11: Total Activity

	(1)	(2)	(3)	(4)	(5)	(6)
	Total Activity	Total Activity	Total Activity	Total Activity	Total Activity	Total Activity
	Low	High	Low	High	Low	High
Post	6.281 (4.678)	10.219* (5.566)	11.111 (8.649)	15.138* (8.462)	47.060*** (17.551)	8.051 (19.969)
HQSXPost	-49.847 (48.696)	-49.366** (22.328)				
High IncomeXPost			-10.590 (10.956)	-7.880 (7.492)		
UrbanizationXPost					-60.019*** (21.439)	2.706 (23.186)
_cons	1799.308*** (631.908)	-1186.199 (840.978)	1791.414*** (631.558)	-1146.019 (847.820)	1891.974*** (641.042)	-1168.096 (843.175)
<i>N</i>	50602	52662	50602	52662	50602	52662

Table 12: Total Activity

	(1) Total Activity	(2) Total Activity	(3) Total Activity	(4) Total Activity	(5) Total Activity	(6) Total Activity
	Low	High	Low	High	Low	High
Post	20.515** (8.109)	9.018 (8.897)	3.006 (4.899)	12.327 (10.559)	5.983 (3.850)	6.520 (6.492)
IndustryXPost	-78.043*** (25.227)	5.473 (31.385)				
HHIAssetsXPost			4.545 (7.526)	-4.137 (18.637)		
ROAXPost					-0.552 (13.603)	-10.983 (101.507)
_cons	1816.018*** (633.489)	-1172.153 (840.927)	1786.836*** (631.483)	-1163.680 (846.977)	1544.044*** (538.146)	-1905.242* (983.204)
<i>N</i>	50602	52662	50602	52662	47264	49183

Table 13: Net Due To

	(1)	(2)	(3)	(4)	(5)	(6)
	Net Due To	Net Due To	Net Due To	Net Due To	Net Due To	Net Due To
	Low	High	Low	High	Low	High
Post	0.033** (0.014)	-0.013* (0.007)	0.036** (0.015)	-0.015 (0.010)	0.032 (0.026)	-0.002 (0.023)
HQSXPost	-0.005 (0.031)	-0.062* (0.033)				
High IncomeXPost			-0.007 (0.005)	0.003 (0.009)		
UrbanizationXPost					0.000 (0.028)	-0.014 (0.029)
_cons	1799.308*** (631.908)	-1186.199 (840.978)	1791.414*** (631.558)	-1146.019 (847.820)	1891.974*** (641.042)	-1168.096 (843.175)
<i>N</i>	50602	52662	50602	52662	50602	52662

Table 14: Net Due To

	(1)	(2)	(3)	(4)	(5)	(6)
	Net Due To	Net Due To	Net Due To	Net Due To	Net Due To	Net Due To
	Low	High	Low	High	Low	High
Post	0.032* (0.016)	-0.011 (0.009)	0.002 (0.018)	-0.028* (0.014)	0.030** (0.015)	-0.010 (0.008)
IndustryXPost	0.005 (0.031)	-0.010 (0.036)				
HHIAssetsXPost			0.046** (0.022)	0.028 (0.023)		
ROAXPost					0.015 (0.017)	0.064 (0.058)
_cons	1799.308*** (631.908)	-1186.199 (840.978)	1791.414*** (631.558)	-1146.019 (847.820)	1891.974*** (641.042)	-1168.096 (843.175)
<i>N</i>	50602	52662	50602	52662	50602	52662

Table 15: Total Lending

	(1) ch_lending	(2) ch_lending	(3) ch_lending	(4) ch_lending	(5) ch_lending
ch_netdueto	0.073 (0.187)	-0.093 (0.227)	-1.210* (0.619)	0.407*** (0.118)	0.989** (0.437)
post		-0.025* (0.015)		-0.004 (0.025)	
ch_netduetoXpost		0.553** (0.229)		-0.293** (0.138)	
cds_5y			-0.000* (0.000)		-0.000 (0.000)
ch_netduetoXcds			0.007*** (0.003)		-0.004* (0.002)
hi_ffXpost				-0.020 (0.027)	
ch_netduetoXhi_ff				-0.590** (0.232)	-2.872*** (0.822)
ch_netduetoXpostXhi_ff				1.061*** (0.300)	
hi_ffXcds					0.000 (0.000)
ch_netduetoXcdsXhi_ff					0.014*** (0.004)
_cons	2.463 (1.695)	1.545 (1.722)	1.725 (1.482)	2.525 (2.441)	1.483 (1.630)
R^2	0.20	0.21	0.21	0.21	0.21
N	96312	96312	96312	96312	96312

Table 16: Total Lending

	(1)	(2)	(3)	(4)	(5)	(6)
	ch_lending	ch_lending	ch_lending	ch_lending	ch_lending	ch_lending
	Low	High	Low	High	Low	High
ch_netdueto	0.330*** (0.076)	0.011 (0.174)	0.439*** (0.107)	-0.153 (0.203)	1.033** (0.438)	-1.524** (0.600)
post			-0.005 (0.025)	0.004 (0.028)		
ch_netduetoXpost			-0.260* (0.135)	0.578*** (0.217)		
cds_5y					-0.000 (0.000)	-0.000 (0.000)
ch_netduetoXcds					-0.004* (0.002)	0.008*** (0.003)
_cons	0.250 (1.834)	7.272* (4.228)	0.102 (2.240)	7.078 (5.610)	0.167 (2.009)	6.164 (4.118)
R^2	0.30	0.27	0.31	0.27	0.31	0.27
N	47253	49059	47253	49059	47253	49059

7. Figures

Figure 1: Internal Liquidity Management

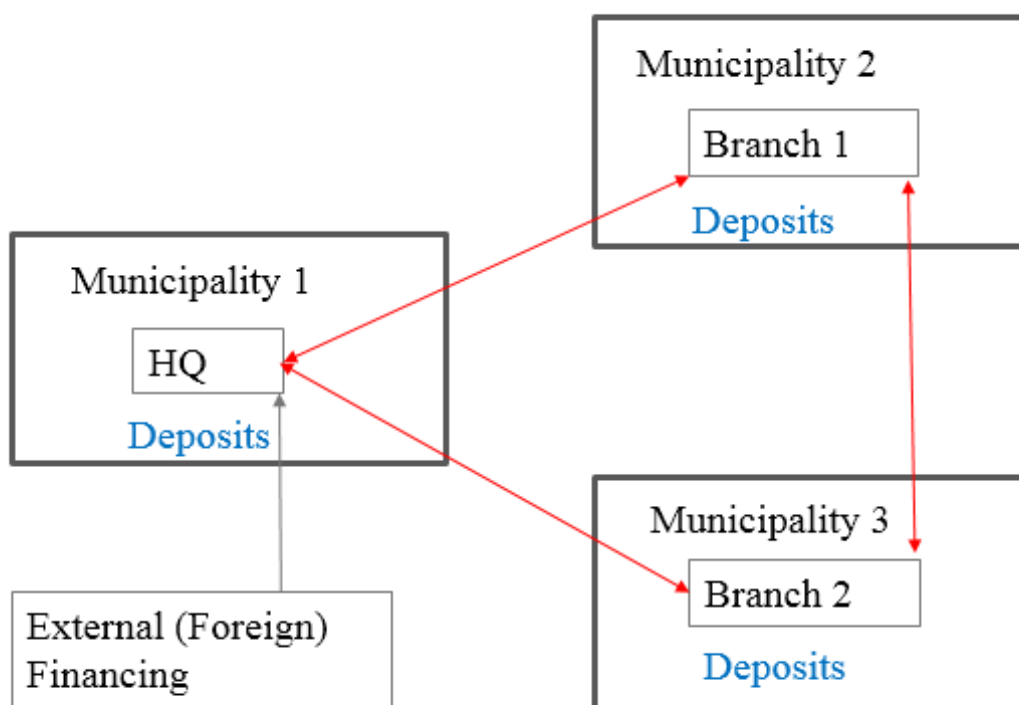


Figure 2: Raw Data: Net Due To/Total Assets vs. per capita Income

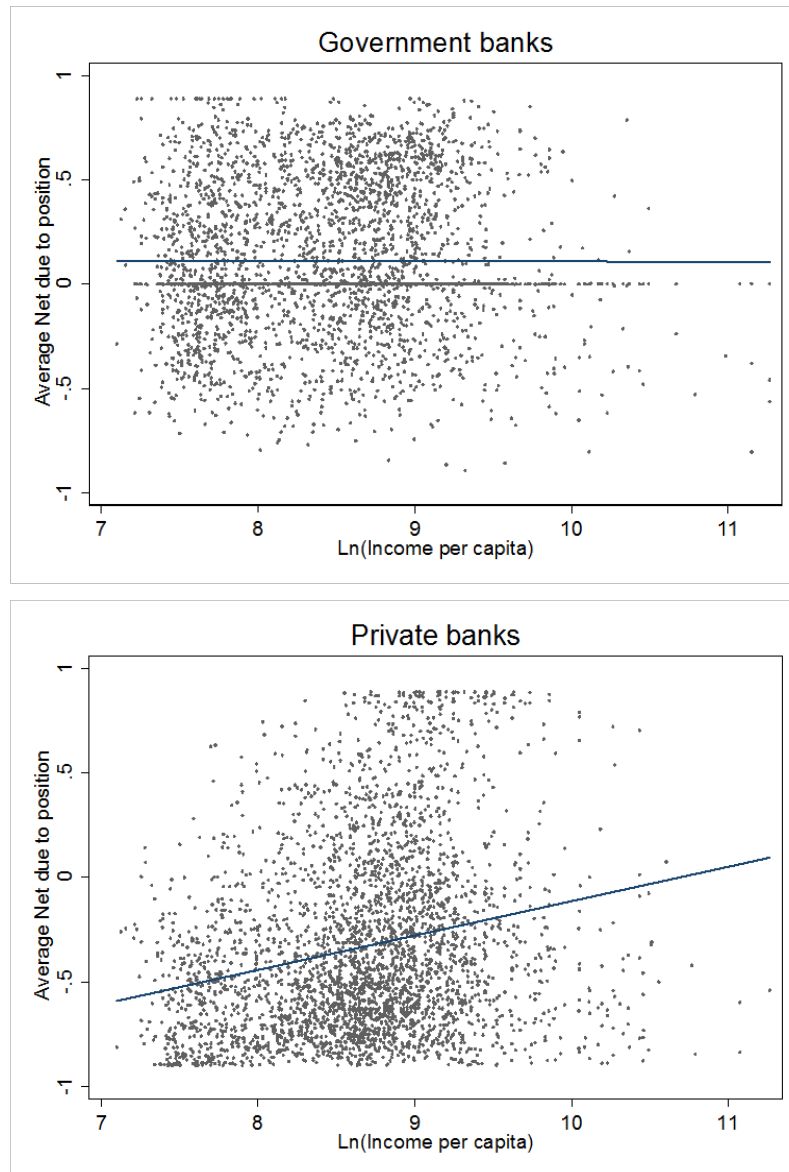


Figure 3: Raw Data: Net Due To/Total Assets vs. Lending

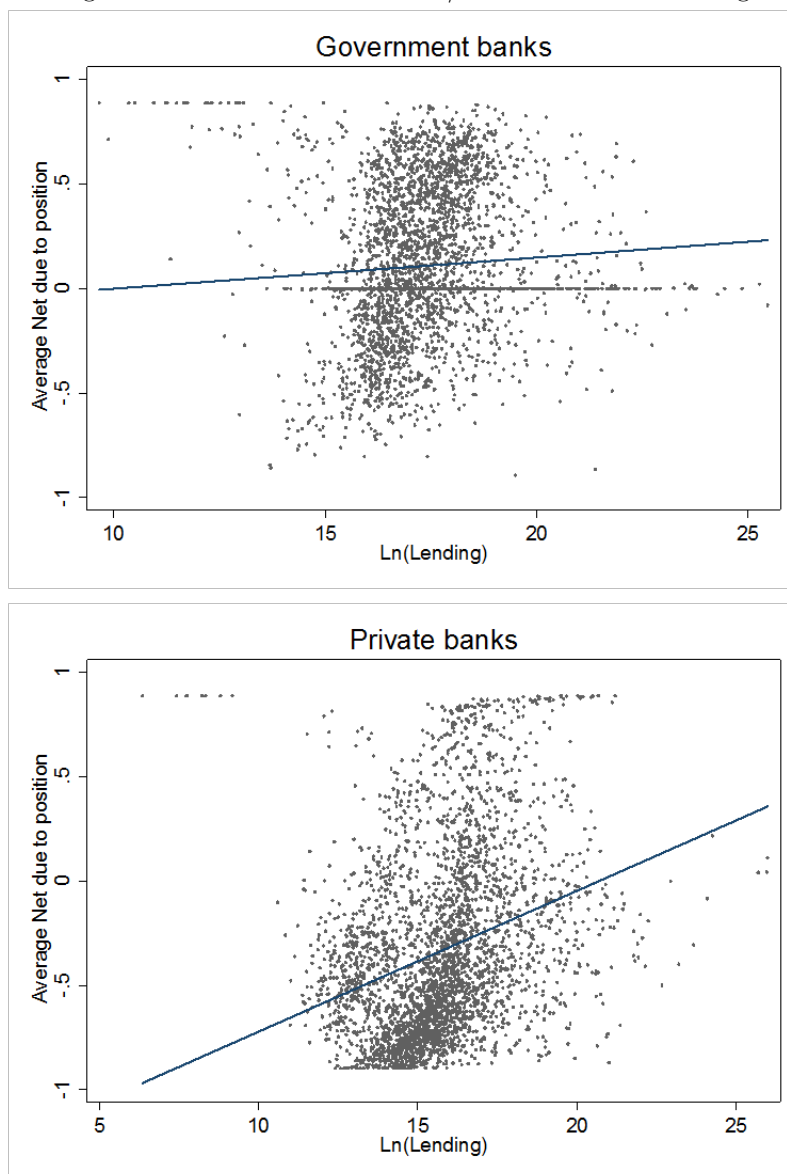
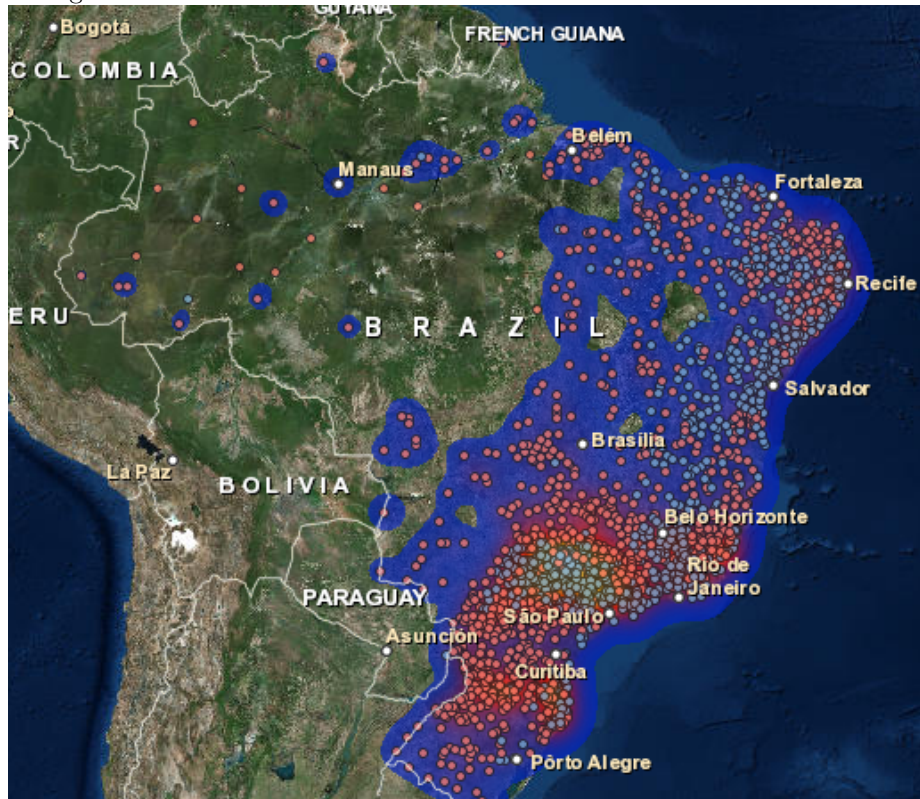


Figure 4: CDS Spreads of Brazilian Banks



Figure 5: Net Lender vs. Borrower Locations of Bank of Brazil Branches



Notes: This map shows which localities are net lenders and which are net borrowers for the Bank of Brazil.

Figure 6: Bank Headquarters



Notes: This map shows the headquarters locations of the banks in our sample.